Lecture-1
19.07.2026

Attendence ———— 10

CT/presentation ———— 20

2টি

Assignment ———— 10

40

Mid ———— 24

final ———— 36

100

প্রথম CT - Lecture-1, 2, 3

Lecture-2
20.07.2026

Basic Discussion

Component of microcomputer System:

There are three parts of computer system.

1. **Central Processing Unit (CPU):** The CPU controls the operation's of the computer and executes all instructions. It is often referred to as the brain of the computer. Two major sub-units carry out through CPU - processing and management.

Summary

For CPU:

⇒ Function: Controls the ~~overt~~ overall operation of the compute

⇒ Execution: Processes and executes all instructions received by the system.

⇒ Role: Serves as the main processing core and is ~~com~~ commonly referred to as the brain of the comp.

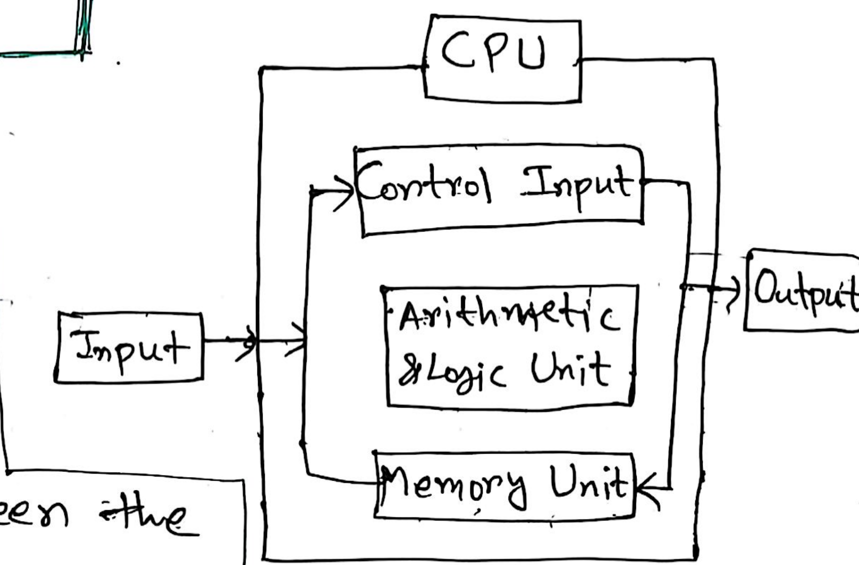
2. Memory:

Memory serves as the main storage unit of the system, retaining the data and instructions required for processing.

⇒ Function: ~~A~~ Serves as the primary storage area within the computer system.

⇒ Role: Holds and stores both data and instructions needed for processing.

3. Input/Output Devices:



Input/Output devices handle System communication by ~~transforming~~ transferring data back and forth between the computer and external equipment or user.

⇒ Function: Enables communication between the computer and external environment.

⇒ Role: Transfer data back and forth between the computer system and external devices.

Q. Why CPU is called the brain of the computer?

⇒ The central processing unit (CPU) is commonly called the brain of the computer for two main reasons based on how it functions within a microcomputer systems:

1. Controls System Operation: It directs and coordinates all operation across the computer, managing how components work together.

2. Executes Instructions: It processes and executes all instructions received by the computer system.

Inside CPU, the Control Unit (CU) manages the ~~fetch~~ fetch fetch-decode-execute cycle, while the ALU carries out actual calculations and data transformation.

Operation of Control Unit

1. ~~Fde~~ Fetch: Control unit ~~fet~~ fetches instructions from memory that need to be executed.

2. Decode: After fetching, control unit decodes the instruction. Decode means understand the binary pattern.

3. Execution: After decoding, control unit executes the instructions.

Operation of Arithmetic Logic Unit (ALU)

1. Arithmetic Operation: Arithmetic operation like Addition, Subtraction, Multiplication etc. is performed here.

2. Logical Operation: Logical operation like AND, OR, NOT is performed here.

3. Data Manipulation: Altering or transforming data to make it more meaningful and manageable.

Microprocessor

Definition: Microprocessor is a processor which incorporates the functions of a CPU on a single Integrated Circuit (IC)

Types: (1) 8086 (2) 8085 (3) 8088

Application: Microprocessors are integrated ~~in~~ circuits designed for big, multi-purpose, general-purpose applications ~~whose~~ where high computing power, flexibility and dynamic multitasking are needed. Examples:

1. Personal Computer: Used as the core processing engine (CPU) for desktops and laptop computers running complex operating systems, Softwares and heavy computational tasks.

2. Mobile Device: Used ~~for~~ in Power smartphones and tablets to handle multi-tasking applications, user interfaces, graphis processing and web browing.

Micro Controller

Definition: Microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system.

Embedded System: Embedded systems are focused on controlling and monitoring specific functions within a devices

Application: Microcontrollers are compact ICs designed for embedded systems Their primary purpose is to execute a single, dedicated task efficiently by utilizing their built-in processor, internal memory, and internal I/O components.

① Washing Machine: Controls and monitors specific ~~dedicated~~ functions such as water levels, wash/spin cycles, timing and temperature.

② **Fire detection and safety Devices:** Monitors sensors coninuously to detect smoke or rapid temperature rises and triggers alarm or safety measures instantly. when thresolds are met.

Difference

Microprocessor	Microcontroller
1. It is used for big applications.	1. It is used to execute a single task
2. It is the heart of computer system	2. It is the heart of the embedded system.
3. It is just a processor. Memory and I/O components have to be connected externally	3. It contains external ~~purposes~~ processor along with internal memory and I/O components
4. Power consumption is high	4. Power consumption is low
5. Cost is more	5. Cost is less
6. Difficult to replace	6. Easy to replace
7. Application: Personal Computer	7. Application: Washing Machine

Microprocessor

Lecture-3

~~31.07~~
26.07.2020

Generation of Microprocessors :

First Generation (betn 1971-1973)

⇒ PMOS technology

⇒ 4 bit processor (16 pins)

⇒ 8 and 16 bit processors (40 pins)

Second Generation (During 1973)

⇒ NMOS technology

⇒ Faster Speed, Higher density

⇒ 4/8/16 bit processors (40 pins).

⇒ Intel 8085 (8 bit processor)

Third Generation (During 1978)

⇒ HCMOS technology

⇒ Faster Speed, Higher Packing density.

⇒ 16 bit Processors (40/48/64 pins)

⇒ Easier to program

⇒ Dynamically relatable programs.

⇒ Intel 8086 (16 bit processor)

Fourth Generation (During 1980)

⇒ HCMOS technology

⇒ 32 bit processors.

⇒ Physical memory space
2^{24} bytes = 16 Mb

⇒ Virtual memory space
2^{40} bytes = 1 Tb.

⇒ Intel 80386

Fifth Generation

Pentium

Pentium (1993)	Pentium Pro (1993)
⇒ 32 bit microprocessor ⇒ 64 bit data bus ⇒ 32 bit address bit ⇒ 4 GB main memory ⇒ 60, 66, 90 MHz	⇒ 32 bit microprocessor ⇒ 64 bit data bus. ⇒ 36 bit address bus. ⇒ 64 GB main memory ⇒ Starts at 150 MHz.
Pentium II (1997)	**Pentium III (1999)**
⇒ 32 bit microprocessor ⇒ 64 bit data bus ⇒ 36 bit address bus ⇒ 64 GB main memory ⇒ Starts at ~~226~~ 266 MHz	⇒ 32 bit microprocessor ⇒ 64 bit data bus ⇒ 36 bit address bus ⇒ 64 GB main memory ⇒ starts at 450 MHz.

Pentium iv (2002)

⇒ 32 bit microprocessor
⇒ 64 bit data bus
⇒ 32 bit address bus
⇒ 4 GB main memory
⇒ starts at 1.4 GHz.

→ A summary table of Generation of Mic.

↳ check <u>Lecture-3</u> (page -8)

<u>8080 / 8085 Microprocessor</u>

⇒ 8 bit microprocessor

⇒ Developed by INTEL in the mid 1970

⇒ NMOS technology

⇒ Address bus 16 bit. $\left\{ 2^{16} = 65536 \text{ byte or } 64\,KB \quad [65536/1024 = 64] \right\}$

⇒ Data bus 8 bit → (Data bus যত bit processor তত bit).

⇒ Non-pipelined processor.

⇒ It has 3 available clock speeds (3 MHz, 5 MHz, 6 MHz)

⇒ 8085 has memory capacity of 64 KB

↳ 2^{16} Bytes $= 2^6 \times 2^{10}$ Bytes

$= 64 \times 1$ KB

$= 64$ KB

$KB = 2^{10}$
$MB = 2^{20}$
$GB = 2^{30}$
$TB = 2^{40}$

⇒ It does not have an instruction queue.

8086 Microprocessor

⇒ 16 bit ~~M~~ microprocessor

⇒ Developed by ~~IN~~INTEL in the year 1978.

⇒ Originally HMOS, now manufactured using HMOS III technique.

⇒ Approximately 29,000 transistors and 40 pin.

⇒ 5V supply.

⇒ 33% duty cycle.

⇒ Address bus 20 bit. $2^{20} = 2 \times 2^{20}$ Bytes

$= 2 \times 1$ MB

$= 2$ ~~MD~~ MB

⇒ Data bus 16 bit

⇒ Pipelined processor

⇒ It has ~~2~~ 3 available clock speeds (5 MHz, 8 MHz and 10 MHz).

⇒ 8086 has memory capacity of 1 MB

⇒ It has 6 byte instruction queue.

*

Q) What is 16-bit Processor: A processor that can process 16-bit ~~data~~ data at a time is called a 16-bit processor.

÷1024

Byte KB MB GB TB

×1024

$\frac{A}{1024^n}$

Difference between 8085 and 8086 Microprocessor

8085	8086
8 bit microprocessor	16 bit microprocessor
Address bus 16 bit	Address bus 20 bit.
Data bus 8 bit	Data bus 16 bit
Non-piplined processor	Pipelined processor
Memory capacity 64 KB	Memory Capacity 1 MB.
NMOS Technology	HMOS III technology

Pipelineing in microprocessors

The process of fetching the next instruction when the present instruction is being executed is called as pipelining.

Types:

Non-pipelined:

Fetch 1	Execute 1	Fetch 2	Execute 2	Fetch 3	Execute

Pipelined:

Fetch 1	Execute 1		
	Fetch 2	Execute 2	
		Fetch 3	Execute 3

Advantages:

a) Requires less time.

b) Speed fast

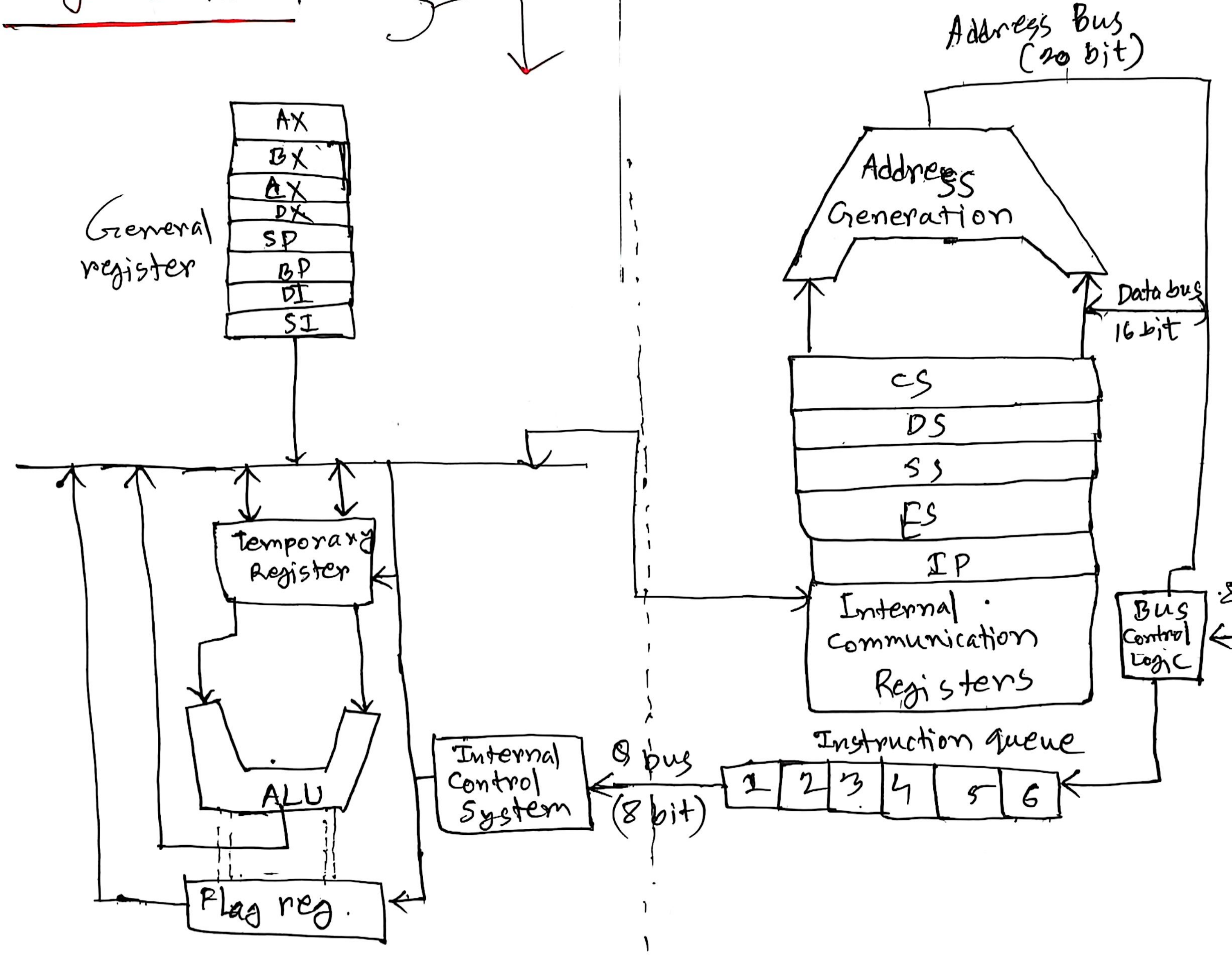
c) Efficiency is more

8086 Microprocessor

Lecture-4

27.07.2026

Internal Block Diagram / Organization:



Bus Interface Unit (BIU)

→ Instruction fetch.

→ Address generation

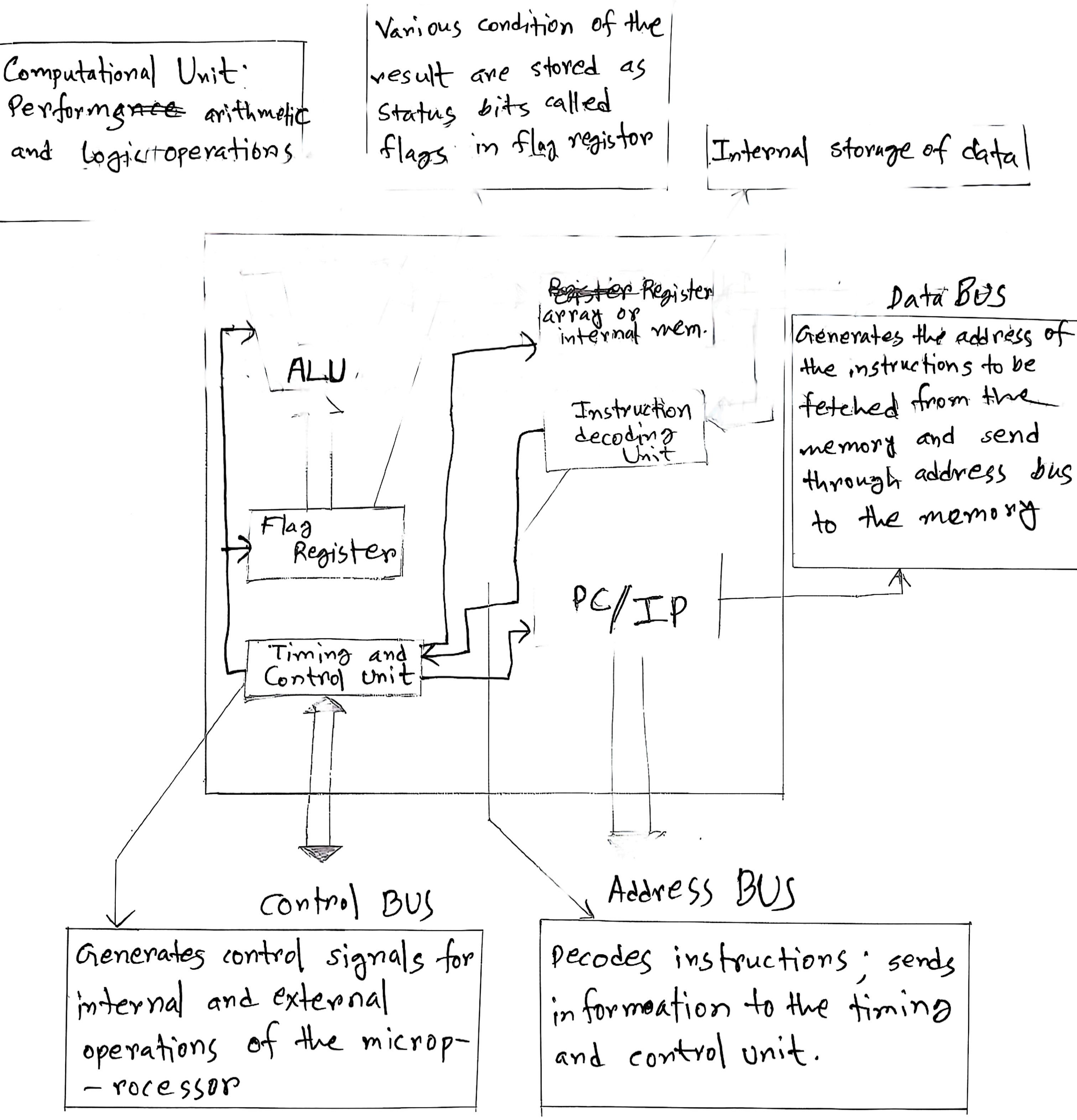
→ Memory and I/o communi--cation.

Execution Unit (EU)

→ Instruction decode and ~~execute~~ execute

→ Arithmetic and logical operation

Functional Block Diagram.



Q. How does queue speed up the processing of the 8086 microprocessor?

⇒ The 8086 microprocessor's 6-byte instruction queue increases speed through instruction prefetching. The BIU fetches instructions and stores them in the queue. At the same time, the EU executes the instructions. Because of instruction fetching and execution overlap, the ~~processor~~ processor's idle time ~~ti~~ is reduced and execution speed increases.

<u>Register of 8086 Microprocessor</u>

Lecture-5
02.08.2026

AX ⟶ AL AH

BX ⟶ BL BH

CX ⟶ CL CH

DX ⟶ DL DH

Data বড় হলে AX

" ছোট " [AL, AH]

আগে AL এ যাবে তারপর AH

low High

<u>এখানের</u>: এই topic <u>Lectur-4 (slide)</u> <u>page(7-20)</u> পড়লেই হবে।

Assembly Language

Lecture-6

03.08.2026

Syntax

~~MO~~

$\underbrace{\text{MOV}}_{\text{opcode}}$ $\underbrace{\text{CX, 5}}_{\text{operand}}$ $\underbrace{\text{; initialize counter}}_{\text{comment}}$

opcode → operation

AX, BX

AX → Destination

BX → source

⇒ Hexa-decimal -এ শেষে অবশ্যই h/H দিতে হবে এবং sequence -এ শুরুর দিকে numerical না থাকলে সামনে 0 দিতে হবে। যেমন:

1FFH | 0 লাগবে না   0ABFF → 0 লাগবে সামনে.

বাকি Lecture-5(slide) page(1-17) পড়তে হবে।